

# Gauge-Invariant Cycle Dynamics of Multi-Inverter Power Networks: Scalarizability, Collective Feedback, and Connectivity–Stability Reversals

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**Abstract**—Scalar graph measures and decoupled impedance models are attractive for screening inverter-dominated power networks, but they discard matrix-valued feedback interactions created by phase-locked loops, voltage/current controls, grid-forming loops, and heterogeneous device dynamics. This paper develops a coordinate-invariant framework for deciding when such a network can be reduced exactly to independent scalar channels and for diagnosing what is lost when it cannot. A matrix-valued network is represented by frequency-dependent edge transports. After a spanning-tree gauge, the non-tree transports are fundamental-cycle holonomies. We prove that all edge transports are globally scalarizable if and only if the fundamental holonomies are simultaneously diagonalizable. A physical energy metric is then used to define a gauge-invariant non-Abelian commutator defect. In 2,000 structured  $2 \times 2$  networks this defect correlates strongly with scalarization error (Spearman  $\rho = 0.8951$ ), while 200 exactly commuting cases remain exact to machine precision. A second counterexample shows that a global unit-return mode can arise collectively even when no individual fundamental cycle is near resonance. We also give an exact two-node MIMO counterexample in which increasing algebraic connectivity destabilizes the relative mode. Finally, an AC-consistent IEEE 39-bus study with reduced synchronous-generator and grid-following-inverter models is used to test the engineering implication. Under nominal PLL tuning, 180 candidate line additions produce no instability. Under an intentionally low-damped PLL stress case, however, four connectivity-improving additions become unstable. For line 25–38, generalized-Fiedler connectivity increases by 0.146% while the full coupled critical mode moves from  $-0.0151 + j17.226$  to  $+0.2502 + j19.779 \text{ s}^{-1}$ ; a decoupled model misses the event. Biorthogonal participation identifies the dominant converter, and retuning one PLL restores stability without removing the line. The results support a narrow conclusion: connectivity is a property of the physical graph, whereas closed-loop stability is a property of the matrix-valued feedback graph.

**Index Terms**—Inverter-dominated power systems, MIMO impedance, graph theory, holonomy, quiver representations, non-Abelian coupling, scalarization, generalized Fiedler vector, PLL, small-signal stability, Braess effect.

## I. INTRODUCTION

MODERN power networks combine synchronous machines, grid-following (GFL) inverters, grid-forming (GFM) inverters, dynamic loads, and controls whose terminal

dynamics are naturally multivariable. In this setting, a line is not merely a scalar weight connecting two otherwise identical oscillators. A perturbation can rotate, amplify, and mix voltage-angle and power channels as it traverses the network. Yet many planning and screening tools remain scalar: Laplacian connectivity, effective resistance, short-circuit-ratio families, or decoupled impedance approximations.

This mismatch matters for two reasons. First, the recently proposed generalized Fiedler vector (GFV) combines network connectivity with nodal inertia and provides a useful notion of *dynamic connectivity* [1]. Second, converter impedance studies have long shown that dropping coupling terms can lead to inaccurate eigenloci or even incorrect stability conclusions when mirror-frequency or  $dq$  coupling is material [6], [7]. These two facts create a practical gap: an upgrade can look favorable to a scalar connectivity metric while the closed-loop MIMO dynamics react in another direction.

Complex-frequency formulations provide an increasingly useful language for heterogeneous power-system dynamics [2], [3], [5]; however, complex frequency and GFV are not the contribution pursued here. Likewise, Braess-type stability degradation after adding a coupling is already known for oscillator networks and power grids [10]. The question addressed here is more specific:

*When does a matrix-valued power-network model admit an exact scalar-channel representation, and what closed-loop information is missing when such a representation does not exist?*

The answer developed in this paper uses cycle algebra. Each local interconnection is treated as a linear map between port spaces. Local state-coordinate changes act as gauge transformations. A spanning-tree gauge removes all tree transports, leaving the fundamental-cycle return maps—their *holonomies*—as the nontrivial global objects. This leads to four contributions.

**C1. Exact scalarizability.** A tree-gauge theorem characterizes when a connected matrix-valued network can be transformed into independent scalar channels: precisely when its fundamental-cycle holonomies are simultaneously diagonalizable.

**C2. Gauge-aware obstruction.** A positive-definite physical port metric yields a coordinate-invariant non-Abelian commutator defect. A 2,000-case structured Monte Carlo study shows that the defect tracks scalarization error,

while commuting networks scalarize exactly to machine precision.

**C3. Collective feedback.** A counterexample shows that global unit-return resonance does not require any single fundamental cycle to be individually close to unit return. This motivates cycle-rooted/trace-based characteristic decompositions rather than a single “bad-loop” margin.

**C4. Connectivity–stability nonordering and mitigation.** An exact minimal MIMO counterexample and an AC-consistent IEEE 39-bus study show that improved scalar connectivity need not improve the full coupled spectrum. In the IEEE 39-bus stress case, a connectivity-improving line creates an instability missed by the decoupled model; targeted PLL retuning restores the margin without undoing the network upgrade.

The paper deliberately separates *scalarizability* from *stability*. Non-Abelianity is not proposed as a stability margin. Rather, it certifies when a scalar reduction cannot retain the complete matrix-valued feedback structure.

## II. RELATED WORK AND POSITIONING

### A. Scalar connectivity and complex frequency

The GFV recently proposed by Pinheiro and Dotta generalizes Fiedler connectivity by incorporating nodal inertia [1]. In parallel, local and system-wide complex-frequency formulations provide a device-agnostic description of voltage magnitude and angle dynamics [2], [3]. Büttner and Hellmann further introduced complex couplings as adaptive dynamical-network variables [5]. These developments provide useful baselines and future observables, but none addresses the exact reducibility of a heterogeneous matrix-valued interconnection into scalar channels.

### B. Coupled versus decoupled converter models

Matrix impedance and generalized Nyquist approaches explicitly retain  $dq$ /sequence coupling. Rygg *et al.* showed that decoupled impedance models are exact only under specific coupling conditions and quantified eigenlocus error [6]. Xin *et al.* likewise showed that neglected coupling can produce incorrect conclusions and proposed a generalized-impedance treatment [7]. The present work is complementary: instead of assessing coupling at one converter, it asks whether the *entire network* admits one global basis in which every interconnection channel decouples.

### C. Cycles, holonomy, and quiver determinants

Cycle-level topological constraints have recently appeared in synchronization through twisted Laplacians and  $U(1)$  holonomy [9]. More generally, Kassel and Lévy derived determinant identities for linear quiver representations in terms of traces of cycle holonomies, including cycle-rooted spanning-forest interpretations for graph vector bundles [8]. We do not claim these algebraic results. We use their perspective to motivate a power-system interpretation: matrix-valued feedback cycles are the coordinate-invariant structures left after trivializing a spanning tree.

### D. Braess effects and stability nonordering

Adding a network coupling can either increase or decrease synchronous-state stability, including in power grids [10]. Our IEEE 39-bus experiment should therefore not be read as a claim that Braess behavior itself is new. The new target is the mismatch between scalar connectivity and a coupled converter mode, together with a scalarizability diagnosis and a controller-retuning remedy.

## III. MATRIX-VALUED CLOSED-LOOP NETWORK

### A. Port reduction and self-energy interpretation

Let  $z$  denote retained terminal variables and  $x_h$  internal machine, converter, filter, and controller states. Linearizing a DAE around an AC solution and eliminating  $x_h$  by a Schur complement yields a frequency-dependent port equation

$$F(s)z = 0, \quad (1)$$

where  $F(s)$  contains the network and the dynamic self-energy of eliminated states. A convenient block splitting is

$$F(s) = D(s) - E(s), \quad (2)$$

with block-diagonal local dynamics  $D(s) = \text{diag}(D_1, \dots, D_n)$  and off-diagonal interconnection blocks  $E_{ij}(s)$ . Whenever  $D(s)$  is invertible,

$$F(s) = D(s)[I - T(s)], \quad T(s) = D^{-1}(s)E(s), \quad (3)$$

so

$$\det F(s) = \det D(s) \det[I - T(s)]. \quad (4)$$

A collective network mode therefore satisfies

$$1 \in \sigma[T(s)]. \quad (5)$$

The block  $T_{ij}(s)$  maps the port space of node  $j$  to that of node  $i$  and embeds inertia, damping, controller memory, and network transfer. This is the point at which the earlier “feedback rheology” interpretation becomes useful: frequency-dependent local dynamics change the transport around network cycles rather than simply adding one scalar damping coefficient.

### B. Gauge transformations

Let each port space be  $V_i \simeq \mathbb{C}^d$  and let  $x_i = Q_i \tilde{x}_i$  with  $Q_i \in \text{GL}(d, \mathbb{C})$ . Then

$$T_{ij} \mapsto \tilde{T}_{ij} = Q_i^{-1} T_{ij} Q_j. \quad (6)$$

The local realization can therefore change while the underlying physical interconnection does not. A useful network descriptor should either be invariant under (6) or carry the physical metric required to transform covariantly.

**Definition 1** (Cycle holonomy). For an oriented closed walk  $C = (i_0, i_1, \dots, i_r = i_0)$ , define

$$H_C = T_{i_0 i_{r-1}} \cdots T_{i_2 i_1} T_{i_1 i_0}. \quad (7)$$

Under (6),  $H_C \mapsto Q_{i_0}^{-1} H_C Q_{i_0}$ . Hence its eigenvalues, trace, determinant, and characteristic polynomial are gauge-invariant.

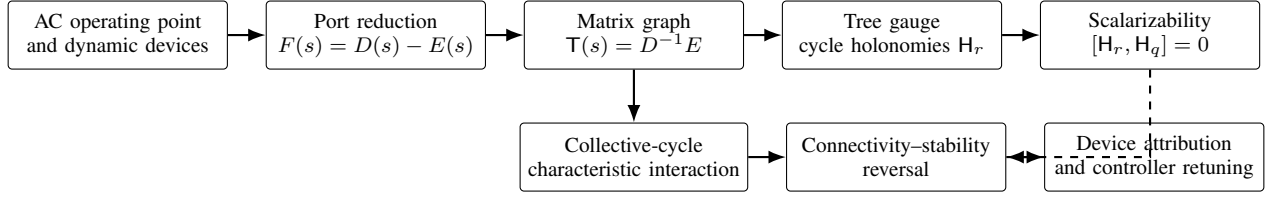


Fig. 1. Paper logic. The contribution is not a new scalar strength metric. The port-reduced MIMO network is treated as a matrix graph; cycle algebra decides exact scalarizability, while the full closed-loop model decides stability.

#### IV. EXACT SCALARIZABILITY

##### A. Single-cycle reduction and tree gauge

The simplest consistency check is a directed cycle with block transports  $T_1, \dots, T_r$ . Let  $T_C$  be the corresponding block cyclic return operator and  $H_C = T_r \cdots T_1$ .

**Lemma 1** (Single-cycle determinant reduction). *For a pure directed cycle of equal block dimension,*

$$\det(I - T_C) = \det(I - H_C). \quad (8)$$

*Proof.* Successive block Gaussian elimination removes the tree-like chain blocks and leaves the Schur complement  $I - T_r \cdots T_1$ . Equivalently, the nonzero closed-walk contribution of the block permutation is the cycle product  $H_C$ .  $\square$

A 3-node,  $2 \times 2$  numerical check gives an absolute determinant mismatch of  $3.5 \times 10^{-16}$ . Under strongly non-unitary local coordinate changes, the spectrum, trace, and determinant of  $H_C$  remain invariant to machine precision. In contrast, the Euclidean quantity  $\sigma_{\min}(I - H_C)$  changed from 0.9026 to 0.5132 under a pure coordinate transformation. This motivates the physical metric introduced below.

##### B. Tree gauge

Choose a spanning tree  $\mathcal{T}_G$  and a root. Assume the tree transports are invertible. Recursively select the  $Q_i$  so that every tree transport becomes the identity. The remaining  $\beta = m - n + 1$  chord matrices are exactly the fundamental-cycle holonomies in this tree trivialization.

**Theorem 1** (Tree-gauge scalarizability). *For a connected matrix-valued network with equal port dimension  $d$  and invertible tree transports, the following statements are equivalent:*

- (i) *there exist local coordinate transformations that make every edge transport diagonal;*
- (ii) *in a tree gauge, all fundamental-cycle holonomies  $H_1, \dots, H_\beta$  are simultaneously diagonalizable.*

*If the  $H_r$  are individually diagonalizable over  $\mathbb{C}$ , (ii) is equivalent to pairwise commutation  $[H_r, H_q] = 0$  for all  $r, q$ .*

*Proof.* In a tree gauge all tree transports are  $I$ . Any further collection of local transformations that preserves all tree identities must satisfy  $S_i = S_j$  on every tree edge; connectedness therefore implies a single similarity  $S$  at all vertices. The only nontrivial transports are the chord matrices  $H_r$ . Hence all edges can be diagonalized if and only if one  $S$  diagonalizes all  $H_r$ . The final statement is the standard simultaneous-diagonalization criterion for a commuting family of diagonalizable complex matrices [15].  $\square$

**Remark 1.** The theorem is a *structural* exactness result. It does not state that a commuting network is stable, only that it admits a representation as  $d$  independent scalar graph channels.

Because  $T_{ij} = T_{ij}(s)$  in a dynamic network, scalarizability can itself be frequency dependent. A network may be nearly Abelian in the electromechanical band yet irreducibly MIMO near a PLL or inner-control resonance. We therefore interpret  $\mathfrak{N}(\omega)$  as a frequency-resolved *model-validity diagnostic*, not as a scalar strength metric.

##### C. Gauge-invariant non-Abelian defect

A Euclidean Frobenius norm of  $[H_r, H_q]$  is not invariant under non-unitary changes of local coordinates. Let  $G \succ 0$  be a physical quadratic port metric, transforming as  $G \mapsto Q^* G Q$ . Define

$$\|A\|_G^2 = \text{tr}(G^{-1} A^* G A) \quad (9)$$

and

$$\mathfrak{N}_{rq} = \frac{\|H_r H_q - H_q H_r\|_G}{\|H_r H_q\|_G + \|H_q H_r\|_G}. \quad (10)$$

**Proposition 1** (Gauge invariance). *The norm (9) and defect (10) are invariant under  $A \mapsto Q^{-1} A Q$ ,  $G \mapsto Q^* G Q$ .*

*Proof.* Substitution into (9), cyclicity of the trace, and  $Q^{-1}(Q^*)^{-1} = (Q^* Q)^{-1}$  cancel all basis factors. Applying the same identity to the commutator and denominator terms yields (10) unchanged.  $\square$

A strongly non-unitary numerical gauge transformation changed  $\mathfrak{N}$  by  $4.2 \times 10^{-17}$ , whereas an unweighted Euclidean singular-value quantity changed materially. This correction is important: gauge invariance requires carrying the physical metric.

#### V. STRUCTURED MONTE CARLO SCALARIZATION TEST

##### A. Protocol

We generated 2,000 four-node networks with a directed spanning tree  $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$  and chords  $2 \rightarrow 0$  and  $3 \rightarrow 0$ . Each  $2 \times 2$  edge transport was a rotated anisotropic positive-definite matrix

$$T_e = R(\theta_e) \text{diag}(d_{e1}, d_{e2}) R^T(\theta_e), \quad (11)$$

with randomized mean gain, anisotropy, and rotation. After tree gauge, the two chords became  $H_1$  and  $H_2$ . We diagonalized  $H_1$  and formed a scalarized model by dropping the remaining off-diagonal entries of  $H_2$ . The scalarization error was the difference between the full and scalarized spectral abscissae of the corresponding block adjacency operators.

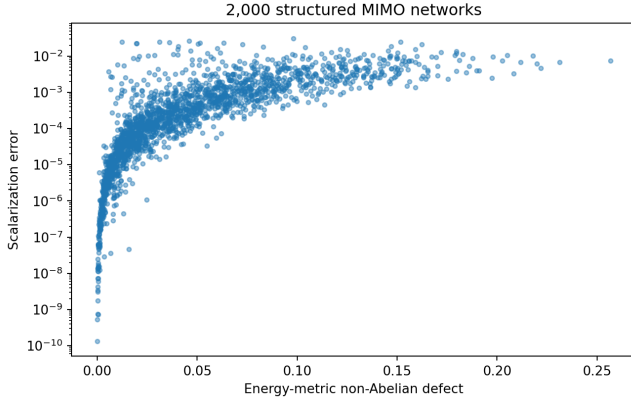


Fig. 2. Structured Monte Carlo test ( $N = 2000$ ). The energy-metric non-Abelian defect increases with the error produced by scalarizing the matrix network. The ordinate is logarithmic.

TABLE I  
SCALARIZABILITY VALIDATION SUMMARY

| Quantity                                       | Result                |
|------------------------------------------------|-----------------------|
| Structured MIMO realizations                   | 2000                  |
| Spearman $\rho_s(\mathfrak{N}, \epsilon_{sc})$ | 0.8951                |
| Pearson $r(\mathfrak{N}, \epsilon_{sc})$       | 0.5306                |
| Upper/lower quartile median error ratio        | $\approx 394$         |
| Exactly commuting gauge tests                  | 200                   |
| Max. commuting-case scalarization error        | $4.8 \times 10^{-15}$ |
| Gauge-change error in $\mathfrak{N}$           | $4.2 \times 10^{-17}$ |

### B. Results

A preliminary one-parameter two-cycle sweep gave a nearly monotone relationship (Spearman  $\rho_s = 0.9983$ ) between a spectral group-commutator metric and scalarization error. The broader structured Monte Carlo campaign gives the more conservative result

$$\rho_s = 0.8951, \quad (12)$$

with Pearson  $r = 0.5306$ . The median scalarization error in the upper quartile of  $\mathfrak{N}$  is approximately 394 times the lower-quartile median. As a separate exactness check, 200 commuting cases were subjected to arbitrary local gauges; the largest scalarization error was  $4.8 \times 10^{-15}$ .

The experiment supports a deliberately limited interpretation:  $\mathfrak{N}$  is an indicator of *global scalarizability*, not a monotone stability index.

## VI. COLLECTIVE-CYCLE FEEDBACK

A tempting cycle criterion would search for one fundamental cycle with  $1 \in \sigma(H_C)$ . The following counterexample shows why that is insufficient.

**Proposition 2** (No single resonant cycle is necessary). *There exist matrix networks for which  $1 \in \sigma(T)$  while  $1 \notin \sigma(H_r)$  for every fundamental-cycle holonomy.*

*Constructive numerical proof.* In a four-port, two-chord tree gauge, the global return operator satisfies  $1 \in \sigma(T)$  to machine precision. However,

$$\sigma(H_1) = \{0.51884, 0.24877\}, \quad (13)$$

$$\sigma(H_2) = \{0.50223, 0.18559\}, \quad (14)$$

so the nearest individual cycle is more than 0.48 from unit return. Nevertheless, sensitivities of the global unit eigenvalue to the two chord gains are 0.14528 and 0.14104, respectively. Thus both cycles participate in the global condition although neither is individually resonant.  $\square$

This motivates a *collective-cycle* characteristic description. Recent trace identities for quiver representations express suitable block determinants through traces of products along cycles, and twisted-Laplacian determinants through cycle-rooted spanning forests [8]. In a power-system implementation, these identities provide a principled route to an attribution of

$$P(s) = \det F(s) = \sum_{\mathcal{F}} \Psi_{\mathcal{F}}(s), \quad (15)$$

where  $\mathcal{F}$  indexes cycle-rooted structures for the applicable lifted representation. Equation (15) is used here as the motivation for future cycle-resolved attribution; the IEEE 39-bus numerical attribution in this draft remains device-level and biorthogonal.

## VII. CONNECTIVITY DOES NOT ORDER MIMO STABILITY

### A. Exact minimal counterexample

Consider

$$\dot{x} = (I \otimes A - L \otimes C)x \quad (16)$$

for two nodes joined by an edge of weight  $w$ , so  $\mu_2(L) = 2w$ . Let

$$A = \begin{bmatrix} 0.719614 & -1.819392 \\ 0.794706 & -1.686641 \end{bmatrix}, \quad C = \begin{bmatrix} 0.283668 & -0.501948 \\ 0.467149 & 0.202568 \end{bmatrix}. \quad (17)$$

Both  $A$  and  $-C$  are Hurwitz. The relative mode is  $A - \mu C$  and

$$\det(A - \mu C) = 0.2919466\mu^2 - 0.9161545\mu + 0.2321517. \quad (18)$$

The two positive roots are

$$\mu_c = 0.2780313, \quad \mu_2^{(+)} = 2.8600578, \quad (19)$$

or, in edge weight,

$$w_c = 0.1390156, \quad w_2 = 1.4300289. \quad (20)$$

The trace remains negative, while the determinant is negative between the two roots. Hence one relative-mode eigenvalue is positive for  $w_c < w < w_2$ . The first crossing at  $w_c$  is the connectivity-increase destabilization used in the argument.

**Theorem 2** (Connectivity–stability nonordering). *There is no universal ordering in which increasing scalar algebraic connectivity  $\mu_2$  must improve the spectral abscissa of a matrix-coupled network. In particular, the system (16)–(17) is stable for  $0 < w < w_c$  and unstable on the nonempty interval  $w_c < w < w_2$ , with  $w_2 = 1.4300289$ , although  $\mu_2 = 2w$  increases monotonically throughout the interval.*

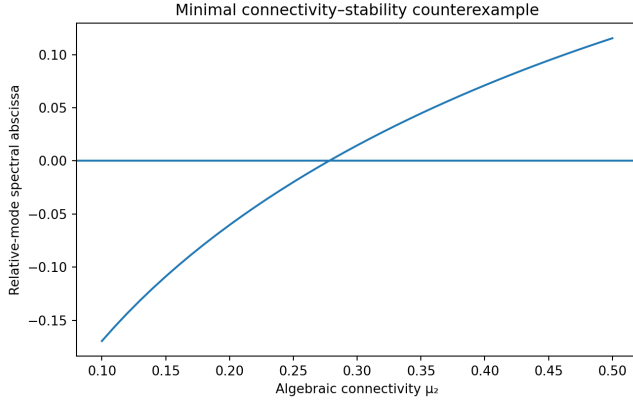


Fig. 3. Exact minimal counterexample. Increasing algebraic connectivity crosses the closed-loop stability boundary at  $\mu_c = 0.2780313$ . The purpose is to disprove a universal connectivity ordering, not to claim a new diffusion-driven instability mechanism.

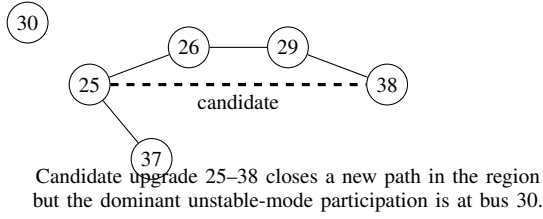


Fig. 4. Local schematic around the IEEE 39-bus candidate intervention. The instability is collective: the endpoint converter at bus 38 participates strongly, but the dominant participating GFL is bus 30.

Related nonmonotone phenomena are known from master-stability and diffusion-driven systems [11]; the contribution here is to use the counterexample as the minimal analytical support for the converter-network screening problem studied next.

## VIII. AC-CONSISTENT IEEE 39-BUS VALIDATION

### A. Electrical benchmark and device placement

The solved MATPOWER New England 39-bus case is used as the base AC operating point [13]. For every candidate new line, the nonlinear AC power flow is re-solved, the full  $P, Q$  injection Jacobian with respect to  $(\theta, V)$  is recomputed, and the 29 load buses are eliminated by a Schur complement. The remaining ten ports correspond to the generator buses.

Four converter placements follow the commonly used low-inertia Config.-II replacement pattern, corresponding here to buses 30, 34, 37, and 38. The public EPFL benchmark provides full-replica configurations in which four synchronous machines are replaced by type-III wind plants [14]. Our present device equations are reduced surrogates rather than the full EPFL Simulink models; this distinction is maintained throughout the paper.

### B. Reduced dynamic models

The synchronous generators use a classical source-behind-reactance representation with swing dynamics. Each GFL

TABLE II  
IEEE 39-BUS DYNAMIC SCREENING CASES

| Case              | PLL $K_p$               | PLL $K_i$ |
|-------------------|-------------------------|-----------|
| Nominal           | 25                      | 400       |
| Low-damped stress | 1                       | 400       |
| Mitigated bus 30  | 3 (bus 30), 1 otherwise | 400       |

device uses a PI PLL and first-order active/reactive current-command surrogates while retaining first-order  $P/Q$  cross-coupling caused by PLL angle error. A filtered  $P$ - $f/Q$ - $V$  GFM helper model is included in the reproducibility code, although the strongest reversal below uses GFL units. The resulting reduced model is physically structured but should not be confused with a vendor-grade EMT realization. The retained core dynamics are, schematically,

$$\dot{\delta}_g = \omega_b \Delta \omega_g, \quad (21)$$

$$M_g \dot{\Delta \omega}_g = -D_g \Delta \omega_g - \Delta P_{e,g}, \quad (22)$$

for each SG, and

$$\dot{\hat{\theta}}_i = -K_{p,i} V_i (\hat{\theta}_i - \theta_i) + K_{i,i} \xi_i, \quad (23)$$

$$\dot{\xi}_i = -V_i (\hat{\theta}_i - \theta_i), \quad (24)$$

for each GFL PLL, augmented by first-order  $P/Q$  command states. Linearizing the terminal rotation gives cross terms proportional to the operating-point  $P_i$  and  $Q_i$ ; these are retained in the full model and removed in the deliberately decoupled baseline. A GFM helper uses filtered  $P$ - $f$  and  $Q$ - $V$  droop states.

For graph baselines, a synchronizing Laplacian

$$K = \sum_{(i,j)} V_i V_j B_{ij} \cos(\delta_i - \delta_j) b_{ij} b_{ij}^T \quad (25)$$

is Kron-reduced to the ten retained ports. Standard Fiedler connectivity is the second eigenvalue of  $K_r$ . GFV connectivity is the second generalized eigenvalue

$$K_r \phi = \mu M \phi, \quad (26)$$

with synchronous-machine inertia and small converter inertia surrogates.

### C. Nominal-tuning falsification gate

With  $K_p = 25$ ,  $K_i = 400$ , the baseline critical mode is

$$-0.15503 + j7.07240 \text{ s}^{-1}. \quad (27)$$

Among 180 candidate line additions at graph distance two or three, 173 increase GFV and 73 both increase GFV and move the critical real part to the right. None becomes unstable. This is an important negative result: the paper does not claim that ordinary line reinforcement generically destabilizes a well-tuned converter network.

#### D. Low-damped PLL stress gate

We next set  $K_p = 1$ ,  $K_i = 400$  on the four GFL units. In the local normalized PLL surrogate this corresponds to approximately  $\omega_n = 20.5$  rad/s (3.26 Hz) and  $\zeta \approx 0.026$ . The baseline is still stable:

$$s_0 = -0.01507 + j17.22591 \text{ s}^{-1}. \quad (28)$$

Of the same 180 candidate lines, four both increase GFV and produce an unstable full coupled mode. Table III lists them.

### IX. CASE STUDY: LINE 25–38

#### A. Connectivity improves while the coupled mode destabilizes

For the full line addition,

$$\mu_2^{\text{GFV}} : 0.2349169 \rightarrow 0.2352597 \quad (+0.1459\%), \quad (29)$$

$$\lambda_2(K_r) : 12.08886 \rightarrow 12.12489 \quad (+0.2980\%). \quad (30)$$

At the same time, the full coupled critical mode moves to

$$s_{\text{crit}} = +0.25019 + j19.77892 \text{ s}^{-1}. \quad (31)$$

The stability crossing occurs already at  $\alpha_c = 0.21298$  of the candidate line admittance; at the crossing, GFV remains 0.1361% above its pre-upgrade value and standard Fiedler 0.3356% above its pre-upgrade value.

#### B. Decoupled model misses the event

A decoupled approximation removes the  $P$ - $V$ ,  $Q$ - $\theta$ , and PLL-induced  $P/Q$  rotation-coupling terms. It predicts

$$-0.15627 + j7.30340 \quad \text{before the line}, \quad (32)$$

$$-0.15625 + j7.30515 \quad \text{after the line}. \quad (33)$$

Thus the scalarized/decoupled representation indicates essentially no adverse change while the full coupled model moves from stable to unstable. This is the main engineering use case for the scalarizability viewpoint: the error is not that a scalar graph metric is numerically poor, but that a scalar representation has omitted a coupled feedback channel.

### X. MODE ATTRIBUTION AND CONTROLLER RETUNING

Biorthogonal participation of the unstable mode is concentrated in the four GFL units, as shown in Fig. 6a. The largest magnitudes are 0.573 at bus 30, 0.380 at bus 37, 0.236 at bus 38, and 0.084 at bus 34. The endpoint bus 38 participates, but the dominant contributor is remote bus 30. This supports the collective-feedback interpretation.

Retuning only the proportional PLL gain at bus 30 gives a zero crossing at

$$K_{p,30}^* = 2.10837. \quad (34)$$

With  $K_{p,30} = 3$ , the new-line case has

$$s_{\text{crit}} = -0.10064 + j19.62628 \text{ s}^{-1}, \quad (35)$$

while the improved graph connectivity is unchanged. The practical workflow is therefore

upgrade  $\rightarrow$  coupled-mode diagnosis  $\rightarrow$  device attribution  $\rightarrow$  retuning  $\rightarrow$  connection is not required for the theorems above and is intentionally left outside the contribution claim. (36)

#### A. Time-domain consistency

A small localized PLL-angle perturbation at bus 38 was propagated through the linearized system. Over 20 s, the unretuned new-line RMS PLL-angle perturbation grows from approximately  $5 \times 10^{-4}$  to  $4.93 \times 10^{-2}$ , whereas the retuned case decays to approximately  $4.1 \times 10^{-5}$ .

### XI. INTERPRETATION: WHAT IS NEW AND WHAT IS NOT

#### A. What the results establish

The experiments and proofs support four statements.

- (1) The exact possibility of a global scalar-channel representation is controlled by simultaneous diagonalizability of fundamental-cycle holonomies after a spanning-tree gauge.
- (2) A physically metric-compatible commutator provides a coordinate-invariant obstruction to that scalarization and is strongly associated with scalarization error in structured MIMO networks.
- (3) Global resonance can be collective across cycles; the absence of an individually resonant fundamental cycle is not sufficient to rule out a unit-return global mode.
- (4) Scalar connectivity and the coupled closed-loop spectrum are not ordered in general. The IEEE 39-bus surrogate provides an AC-consistent converter-network example in which GFV and Fiedler both improve while a coupled PLL mode destabilizes, and targeted retuning repairs the instability without undoing the upgrade.

#### B. What the results do not establish

The non-Abelian defect  $\mathfrak{N}$  is not claimed to be a stability margin. High  $\mathfrak{N}$  means that a scalar representation is structurally questionable, not that the system is necessarily unstable. The nominal IEEE 39-bus scan found no unstable line addition. The reversal appears in a deliberately low-damped PLL stress case, which is exactly where coupled modeling should matter most.

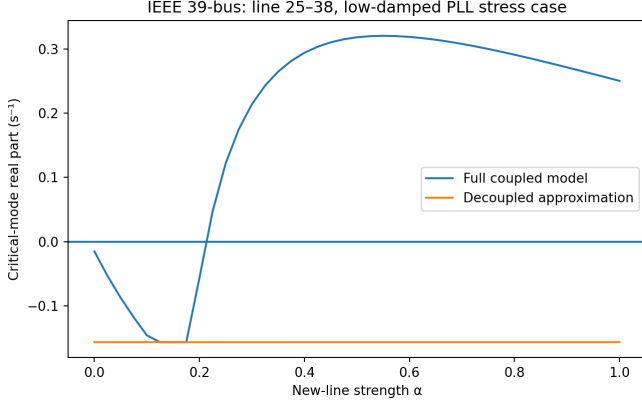
The AC network and operating-point re-solution are benchmark-consistent, but the GFL/GFM dynamic blocks are reduced surrogates. Therefore the IEEE 39-bus result is a constructive AC-consistent proof-of-concept, not yet a full-fidelity industry validation.

#### C. Relation to complex frequency

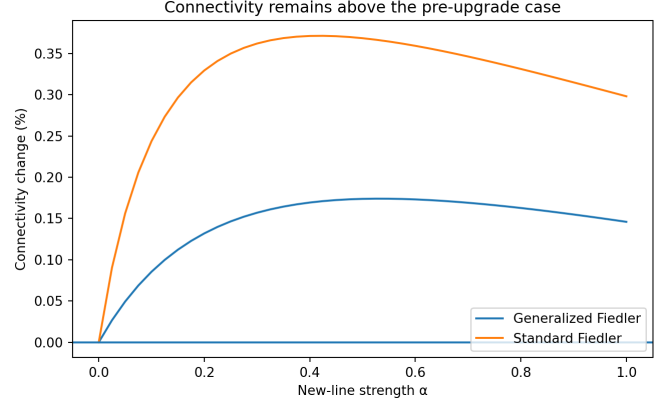
Sofos and Milano showed that complex frequency can coincide with system eigenvalues for diagonalizable LTI systems under the appropriate transformed states, while nonlinear trajectories generally break this equivalence [3]. This gives a natural future observation layer: the coupled mode predicted by the matrix graph may be tracked through complex-frequency patterns measured at converter terminals. That measurement connection is not required for the theorems above and is intentionally left outside the contribution claim.

TABLE III  
CONNECTIVITY-IMPROVING LINE ADDITIONS THAT DESTABILIZE THE LOW-DAMPED PLL STRESS CASE

| Line  | Graph distance | $\Delta\text{Fiedler}$ (%) | $\Delta\text{GFV}$ (%) | New real part ( $\text{s}^{-1}$ ) | Imag. part ( $\text{s}^{-1}$ ) | Interpretation     |
|-------|----------------|----------------------------|------------------------|-----------------------------------|--------------------------------|--------------------|
| 25–38 | 3              | 0.298                      | 0.146                  | 0.2502                            | 19.7789                        | strongest reversal |
| 17–29 | 3              | 4.419                      | 0.740                  | 0.0963                            | 19.8752                        | unstable           |
| 2–29  | 3              | 0.381                      | 0.952                  | 0.0252                            | 19.6046                        | unstable           |
| 4–30  | 3              | 2.013                      | 1.020                  | 0.00057                           | 19.7921                        | near boundary      |

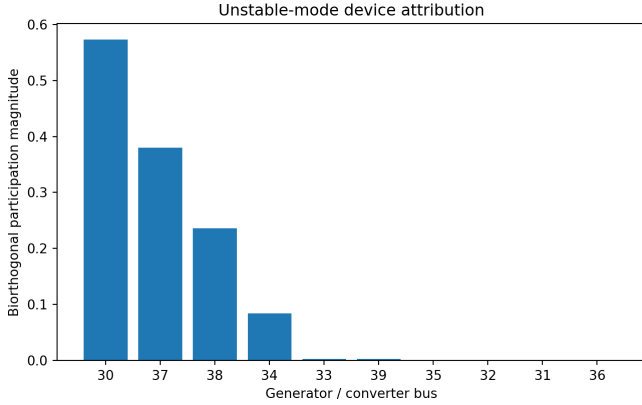


(a) Full coupled versus decoupled critical real part.

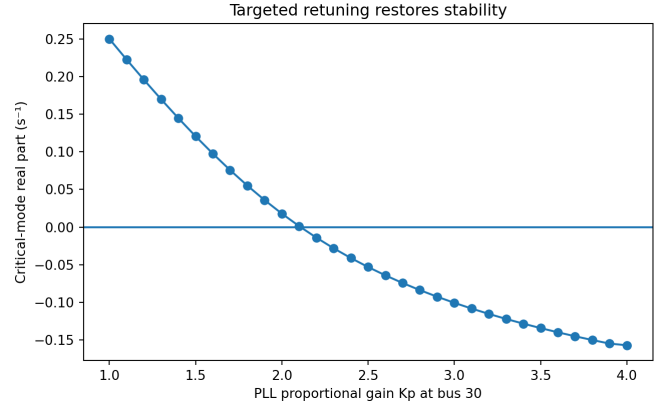


(b) Standard and generalized-Fiedler connectivity changes.

Fig. 5. IEEE 39-bus line 25–38 stress-case sweep. Connectivity remains above the pre-upgrade case while the full coupled model crosses into instability. The decoupled approximation does not reveal the dangerous mode.



(a) Biorthogonal device participation of the unstable mode.



(b) Retuning only  $K_{p,30}$  restores the stability margin.

Fig. 6. Diagnosis and mitigation. The line upgrade is retained; only the dominant converter controller is retuned.

## XII. ENGINEERING WORKFLOW

The proposed use of the framework is not to replace full MIMO stability analysis. It is to decide when a reduced scalar analysis is defensible and, when it is not, to preserve the smallest coupled model required for diagnosis.

- Step 1.** Solve the AC operating point and obtain a local port model  $F(s)$ , either analytically, by frequency response, or by DAE elimination.
- Step 2.** Construct the matrix return graph  $T(s)$  and a fundamental-cycle basis.
- Step 3.** Carry a physical port metric and evaluate  $\Re(\omega)$  over the frequency band of interest.

- Step 4.** If the fundamental holonomies commute (within tolerance) and are diagonalizable, use the exact or near-exact scalar channels. Otherwise retain the coupled model.
- Step 5.** Compare graph planning metrics such as GFV with the full coupled spectral margin; do not infer one from the other.
- Step 6.** If an intervention creates a dangerous coupled mode, use left/right participation and, ultimately, cycle-rooted attribution to identify the controller-network structures that move it.
- Step 7.** Retune the smallest responsible controller set and verify the full coupled and time-domain response.

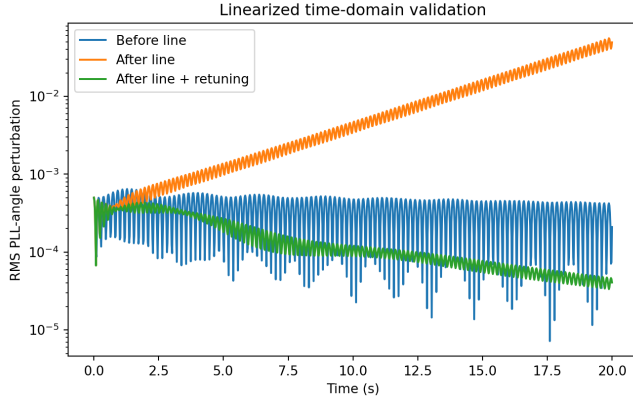


Fig. 7. Linearized time-domain validation of the line 25–38 stress case. The new-line response grows without retuning and decays after targeted PLL retuning.

### XIII. LIMITATIONS AND FINAL VALIDATION GATE

A journal submission should add one full-order replication. The public EPFL IEEE 39-bus extensions provide detailed low- and very-low-inertia Simulink models [14]; alternatively, ANDES/EMT implementations may be used. The required gate is narrow:

- (i) reproduce a connectivity-improving network intervention in a full DAE/EMT model;
- (ii) compare full MIMO and decoupled analysis;
- (iii) confirm the critical mode in time domain;
- (iv) retune the highest-participation controller and recover the margin without undoing the network upgrade.

A successful full-order reversal would elevate the current proof-of-concept into a strong validation of the proposed screening logic. A failure to reproduce it would still leave the scalarizability theorem and nonordering counterexample intact, while narrowing the engineering claim.

### XIV. CONCLUSION

This paper separates two questions that are often mixed in inverter-dominated network analysis: how well connected the physical network is, and whether its matrix-valued closed-loop interactions can be represented by independent scalar channels. The first can be described by Fiedler-type metrics; the second is governed by the algebra of dynamic cycle transports. A tree-gauge theorem gives an exact scalarizability condition, and an energy-metric non-Abelian defect provides a coordinate-invariant measure of departure from that condition. Monte Carlo results show that this departure tracks scalarization error. Counterexamples further show that global resonance can be collective across cycles and that algebraic connectivity does not order MIMO stability. In the AC-consistent IEEE 39-bus stress study, a line addition improves both standard and generalized Fiedler connectivity while destabilizing a coupled PLL mode that a decoupled model misses; targeted retuning restores stability without removing the line. The resulting message is deliberately narrow: *connectivity belongs to the physical graph; stability belongs to the closed-loop matrix graph.*

## APPENDIX A

### ADDITIONAL PROOF DETAILS FOR THEOREM 1

Let the spanning tree be rooted at  $r$ . Set  $Q_r = I$ . For every tree edge  $i \rightarrow j$ , define  $Q_j = T_{ji}Q_i$ ; if the stored orientation is opposite to the tree traversal, use the inverse transport. Then

$$Q_j^{-1}T_{ji}Q_i = I. \quad (37)$$

Every chord  $a : u \rightarrow v$  becomes  $Q_v^{-1}T_aQ_u$ , which is the holonomy obtained by traversing the tree from  $r$  to  $u$ , then the chord, then the inverse tree path from  $v$  to  $r$ . If a second gauge  $S_i$  leaves every tree edge equal to  $I$ , then  $S_j^{-1}S_i = I$  on every tree edge; connectedness implies  $S_i = S$  for all  $i$ . Consequently all chords can be made diagonal exactly when one common  $S$  diagonalizes them. This proves the equivalence in Theorem 1.

## APPENDIX B

### COUNTEREXAMPLE STABILITY THRESHOLD

For (17), the relative-mode trace is

$$\text{tr}(A - \mu C) = \text{tr}(A) - \mu \text{tr}(C), \quad (38)$$

which remains negative at the first positive determinant root. Hence the two-dimensional Routh–Hurwitz conditions reduce the first loss of stability to  $\det(A - \mu C) = 0$ , yielding (20). This makes the counterexample exact rather than simulation-dependent.

## APPENDIX C

### COLLECTIVE-CYCLE COUNTEREXAMPLE DATA

For Proposition 2, the two chord holonomies in tree gauge are

$$H_1 = \begin{bmatrix} 0.455247 & 0.113039 \\ 0.116163 & 0.312369 \end{bmatrix}, \quad H_2 = \begin{bmatrix} 0.498997 & 0.057555 \\ 0.017625 & 0.188825 \end{bmatrix}. \quad (39)$$

Their energy-metric commutator defect for  $G = I$  is 0.08588. The full  $8 \times 8$  return operator is scaled so that its dominant eigenvalue is exactly one; the complete matrix is included in the reproducibility package. The important point is structural: the global unit-return condition emerges without either  $H_1$  or  $H_2$  approaching one individually.

## APPENDIX D

### REDUCED DEVICE-MODEL SCOPE

The IEEE 39-bus surrogate intentionally retains only the dynamic mechanisms required to test the paper hypothesis. SG inertia/damping is represented explicitly; GFL devices retain PLL states and the operating-point rotation that mixes active/reactive injection channels; line additions trigger a new nonlinear AC solution before each linearization. Current limits, dc-link energy, detailed electromagnetic filters, saturation, protection logic, and vendor-specific inner loops are omitted. These omissions define the final full-order validation gate rather than being hidden assumptions.

## APPENDIX E

### REPRODUCIBILITY SUMMARY

The numerical package accompanying this draft contains: (i) the 2,000-case scalarizability Monte Carlo script and CSV; (ii) the analytical counterexample script and sweep; (iii) the MATPOWER case39 data; (iv) the AC power-flow, Schur-reduction, SG/GFL/GFM surrogate, line-screening, participation, retuning, and time-response scripts; and (v) all figure data. The random seed for the scalarizability campaign is 20260823.

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